

Experiment – 12

Develop a mobile application to send an email

AIM:

To develop a mobile application that allows the user to send an email using the app. The application provides an interface to enter the recipient's email address, subject, and message content, and uses Android's built-in email functionality to send the email.

ALGORITHM:

1. Initialize the Application:

- *Create a new Android project in Android Studio.*
- *Create a MainActivity for handling the user interface.*

2. Design the User Interface:

- *Add the following components in the XML layout:*
 - *EditText for recipient email address*
 - *EditText for subject*
 - *EditText for message*
 - *Button to send email*
 - *Toast/TextView for feedback*

3. Declare Variables:

- *Declare variables for EditText fields and Button.*
- *Store email, subject, and message in String variables.*

4. Set Button Listener:

- *Attach OnClickListener to the Send button.*
- *Retrieve user input using getText().toString().*

5. Validate Inputs:

- *Check if email, subject, and message fields are not empty.*

- *Validate email format.*
- *Show Toast message if validation fails.*

6. Create Intent:

- *Use Intent to open email applications.*
- *Set action as Intent.ACTION_SEND.*
- *Add email details using intent extras.*

7. Attach Files (Optional):

- *Use Intent.EXTRA_STREAM for file attachments (if required).*

8. Send Email:

- *Use startActivity() to open email client.*
- *If no email app is found, show error message.*

9. Display Feedback:

- *Show success or failure messages using Toast.*

10. End Process:

- *Allow user to send another email or exit the app.*

CODE:

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
```

```
<LinearLayout
```

```
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:orientation="vertical"
    android:padding="16dp"
    android:layout_width="match_parent"
    android:layout_height="match_parent">
```

```
<EditText
```

```
    android:id="@+id/email"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
```

```
    android:hint="Recipient Email"
    android:inputType="textEmailAddress" />
```

```
<EditText
    android:id="@+id/subject"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:hint="Subject"
    android:inputType="text" />
```

```
<EditText
    android:id="@+id/message"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:hint="Message"
    android:inputType="text"
    android:lines="5" />
```

```
<Button
    android:id="@+id/sendBtn"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Send Email" />
```

```
</LinearLayout>
```

MainActivity.java

```
package com.example.emailappilcation;
```

```
import android.content.Intent;
import android.os.Bundle;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;
```

```
import androidx.appcompat.app.AppCompatActivity;
```

```
public class MainActivity extends AppCompatActivity {
```

EditText email, subject, message;
Button sendBtn;

@Override

```
protected void onCreate(Bundle savedInstanceState) {  
    super.onCreate(savedInstanceState);  
    setContentView(R.layout.activity_main);
```

```
  
    email = findViewById(R.id.email);  
    subject = findViewById(R.id.subject);  
    message = findViewById(R.id.message);  
    sendBtn = findViewById(R.id.sendBtn);
```

```
    sendBtn.setOnClickListener(v -> {
```

```
        String to = email.getText().toString().trim();  
        String sub = subject.getText().toString().trim();  
        String msg = message.getText().toString().trim();
```

```
        if (to.isEmpty() || sub.isEmpty() || msg.isEmpty()) {  
            Toast.makeText(MainActivity.this,  
                "Please fill all fields",  
                Toast.LENGTH_SHORT).show();  
            return;  
        }  
    }
```

```
        Intent intent = new Intent(Intent.ACTION_SEND);  
        intent.setType("message/rfc822");  
        intent.putExtra(Intent.EXTRA_EMAIL, new String[]{to});  
        intent.putExtra(Intent.EXTRA_SUBJECT, sub);  
        intent.putExtra(Intent.EXTRA_TEXT, msg);
```

```
        try {  
            startActivity(Intent.createChooser(intent, "Send Email"));
```

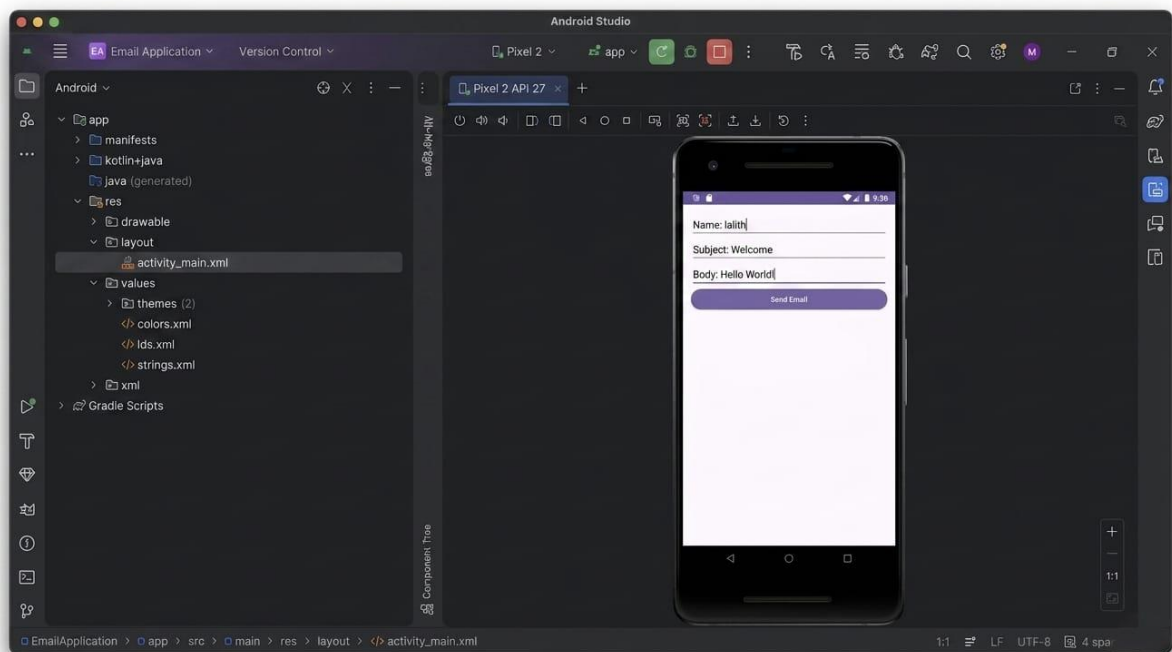
```
        }  
        Toast.makeText(MainActivity.this,  
            "Email app opened successfully",  
            Toast.LENGTH_SHORT).show();
```

```

    } catch (Exception e) {
        Toast.makeText(MainActivity.this,
            "No email app found",
            Toast.LENGTH_SHORT).show();
    }
});
}
}
}

```

OUTPUT:



RESULT:

A mobile application was successfully developed to send an email using Android Studio.